Yujun Zhou

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Summary

I am a Ph.D. student at the University of Notre Dame, advised by Prof. Xiangliang Zhang, graduating in **Spring 2027**. My research includes LLM reasoning and reinforcement learning, long-horizon coding agents, and reliable evaluation and alignment. Previously, I interned at Microsoft and Tencent AI Lab.

Seeking U.S. full-time research roles starting Spring/Summer 2027.

Education

2023 - Present	PhD in Computer Science at **University of Notre Dame, USA**	(GPA: 4.00/4.0)
2021 - 2022	Master's Degree in Computer Science at **KAUST, Saudi Arabia**	(GPA: 3.83/4.0)
2017 - 2021	Bachelor's Degree in Computer Science at **UESTC, China**	(GPA: 3.92/4.0)

Selected Publications

* Equal contribution.

[1] **Yujun Zhou**, Jingdong Yang, Yue Huang, Kehan Guo, Zoe Emory, Bikram Ghosh, Amita Bedar, Sujay Shekar, Zhenwen Liang, Pin-Yu Chen, Tian Gao, Werner Geyer, Nuno Moniz, Nitesh V. Chawla, and Xiangliang Zhang. "Benchmarking large language models on safety risks in scientific laboratories". ***Nature Machine Intelligence*** 8(1), 20–31 (2026). doi:10.1038/s42256-025-01152-1.

[2] Chuan Zou*, **Yujun Zhou***, Xiaonan Luo, and Xiangliang Zhang. "StateHarness: Maintaining Evolving Project State for Long-Horizon Coding Agents". *arXiv preprint*, 2026. arXiv:2610.17616.

[3] **Yujun Zhou**, Kehan Guo, Haomin Zhuang, Xiangqi Wang, Yue Huang, Zhenwen Liang, Pin-Yu Chen, Tian Gao, Nuno Moniz, Nitesh V. Chawla, Xiangliang Zhang. "TellOnce: Correcting Coding Agents Once, Improving Their Future Actions". In *arXiv preprint*. arXiv:2606.13174. [Code]

[4] **Yujun Zhou***, Yue Huang*, Han Bao*, Kehan Guo, Zhenwen Liang, Pin-Yu Chen, Tian Gao, Werner Geyer, Nuno Moniz, Nitesh V Chawla, Xiangliang Zhang. "Alignment Risks from Capability-Seeking RL Training". In: ***ICML***. 2026.

[5] **Yujun Zhou***, Zhenwen Liang*, Haolin Liu, Wenhao Yu, Kishan Panaganti, Linfeng Song, Dian Yu, Xiangliang Zhang, Haitao Mi, Dong Yu. "Evolving Language Models without Labels: Majority Drives Selection, Novelty Promotes Variation". In *arXiv preprint*. arXiv:2509.15194.

[6] **Yujun Zhou***, Jiayi Ye*, Zipeng Ling*, Yufei Han, Yue Huang, Haomin Zhuang, Zhenwen Liang, Kehan Guo, Taicheng Guo, Xiangqi Wang, and Xiangliang Zhang. "Dissecting Logical Reasoning in LLMs: A Fine-Grained Evaluation and Supervision Study". In: ***EMNLP Findings***. 2025.

[7] **Yujun Zhou**, Yufei Han, Haomin Zhuang, Kehan Guo, Zhenwen Liang, Hongyan Bao, and Xiangliang Zhang. "Defending Jailbreak Prompts via In-Context Adversarial Game". In: ***EMNLP***. 2024.

[8] **Yujun Zhou***, Yufei Han*, Haomin Zhuang, Hongyan Bao, and Xiangliang Zhang. "Attack-free Evaluating and Enhancing Adversarial Robustness on Categorical Data". In: ***ICML***. 2024.

[9] Zhenwen Liang*, **Yujun Zhou***, Sidi Lu, Xiangliang Zhang, Haitao Mi, Dong Yu. "Too Correct to Learn: Reinforcement Learning on Saturated Reasoning Data". In **ACL**. 2026.

[10] Zhenwen Liang*, Ruosen Li*, **Yujun Zhou***, Linfeng Song, Dian Yu, Xinya Du, Haitao Mi, Dong Yu. "Your Reasoning Model is Secretly a Reward Model - Optimization-Free Verification from Experience". In **ACL**. 2026.

[11] Xiangqi Wang, Yue Huang, Yanbo Wang, Xiaonan Luo, Kehan Guo, **Yujun Zhou**, Xiangliang Zhang. "AdaReasoner: Adaptive Reasoning Enables More Flexible Thinking". In ***NeurIPS***. 2025 (Spotlight).

Research Experience

LLM Reasoning and Reinforcement Learning

- **EVOL-RL: Label-Free Reasoning RL.** Developed an evolution-inspired framework combining majority-based selection with novelty rewards to mitigate entropy collapse, preserve reasoning diversity, and improve generalization without labeled rewards.
- **FineLogic: Logical Reasoning Evaluation and Supervision.** Evaluated accuracy, stepwise soundness, and representation alignment; compared four supervision formats to study their generalization to logical reasoning.

Long-Horizon Coding Agents

- **StateHarness: Evolving Project State.** Studied runtime support for maintaining project state across long-horizon coding tasks, including findings, decisions, repository relationships, and unfinished work, without updating model parameters.
- **TellOnce: Learning from User Corrections.** Developed a personalization layer that learns persistent, scoped, and revisable behavioral rules from conversational feedback, using reusable checks when reliable verification is possible. Evaluated transfer to subsequent tasks without further user corrections.

Trustworthy Evaluation and Alignment

- **Alignment Risks from Capability-Seeking RL Training.** Investigated how capability-seeking RL can induce exploitation of training-time vulnerabilities and deceptive behaviors that mimic genuine performance gains.
- **LabSafety Bench.** Developed a benchmark for evaluating LLM reliability and safety awareness in scientific laboratories, extending evaluation to safety-critical real-world scenarios.
- **In-Context Adversarial Game: Adversarial Co-Evolution.** Developed a training-free framework in which attack and defense agents iteratively adapt through interaction. Converted feedback-validated reflections into a persistent set of reusable defense rules, incrementally adding, revising, reinforcing, and removing insights to guide subsequent behavior.

Industry Experience

Microsoft — Research Intern May–Aug. 2026

- **Enterprise AI Benchmarking and Trustworthy LLM Evaluation.** Built realistic enterprise-oriented benchmarks for LLM agents, focusing on evidence-aware reasoning, reliable decision-making, and trustworthy behavior in complex, long-horizon workplace scenarios.

Tencent AI Lab (Tencent America LLC) — Research Intern May–Dec. 2025

- **LLM Reasoning and Reinforcement Learning.** Researched label-free RL and mathematical reasoning in the AGI (Audio/Speech/Language & Multimodal) team, including EVOL-RL, reasoning diversity, and methods for expanding problem-solving capabilities.

Services

Workshop Organizer: RelSciFM Workshop, KDD 2026.
Invited Conference/Journal Reviewer: ICLR, NeurIPS, ARR, COLM, AAAI, KDD, CVPR, ICML